

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

TRAINING TR-102 REPORT DAY 21

21 JULY 2025

Overview:

Today's session marked the beginning of my hands-on AI project focused on Face Recognition. I explored the core idea behind the system — recognizing and verifying faces using machine learning and computer vision. The project aims to replicate real-life face-based systems such as biometric attendance, device unlocks, and surveillance monitoring.

The day involved understanding how face recognition works in the background — from capturing an image, detecting faces, extracting features, and finally comparing those features using a model to recognize individuals. I got familiar with libraries, discussed dataset planning, and the necessary installations to run the project smoothly.

Learning Objectives:

- Understand the concept and purpose of a Face Recognition system.
- Explore real-life applications and significance in today's digital world.
- Learn about key Python libraries used in face recognition (like OpenCV, face recognition, NumPy).
- Set up the project environment using Visual Studio Code.
- Install dependencies and ensure camera access for testing.
- Learn best practices for project folder structure and dataset organization.

Project Setup

I began with creating a virtual environment and installing necessary libraries:

```
python -m venv myenv
.\myenv\Scripts\activate
pip install opencv-python face_recognition numpy
```

After that, I verified camera access and configured VS Code for live testing. I also created structured folders for dataset, encodings, and output to streamline the process.

A sample Python script was written to test if our webcam could capture frames:

```
import cv2

cap = cv2.VideoCapture(0)
while True:
    ret, frame = cap.read()
    cv2.imshow('Webcam Test', frame)
    if cv2.waitKey(1) & 0xFF == ord('q'):
        break
cap.release()
cv2.destroyAllWindows()
```

Tools & Libraries Used

- Python 3.10
- OpenCV – For capturing images and face detection.
- face recognition – For feature encoding and face comparison.
- NumPy – For handling numerical arrays.
- Visual Studio Code – IDE for writing and executing code.
- Virtual Environment – To isolate project dependencies.
- Other Tools: pip, virtual environment, GitHub

Key Highlights of Project Setup & Initialization

In this phase of the face recognition project, I clearly understood the fundamental difference between face detection (locating a face in an image) and face recognition (identifying whose face it is). To ensure smooth development, I set up a virtual environment and installed essential dependencies using pip, including libraries like OpenCV, NumPy, and face recognition. Lastly, a basic webcam test script using OpenCV was executed to verify that the system and environment were properly configured and capable of capturing live video input for further processing.

Conclusion:

Today's session is the foundation for the face recognition project. With the project environment ready and our understanding sharpened, we are now equipped to move into the dataset collection and face encoding phase. The real-world relevance of this project makes it an exciting opportunity to implement machine learning and computer vision skills practically.